7COSC013W: Foundations of AI - Coursework 2
A Critical, Historical, Ethical, Legal, and Social Analysis Report

Forecast-Driven Vehicle Routing for Bike-Share Rebalancing
(A Case Study of the Jersey City & Hoboken Citi Bike Network)

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Introduction

Coursework 1 focused on creating and deploying a hybrid AI system to handle bike-share rebalancing for the Citi Bike network in Jersey City and Hoboken, which combined machine learning for demand forecasting alongside search and optimization for vehicle routing. While the project concentrated on developing and evaluating a functional technical solution, this study takes a critical look at it, locating it within the larger history, theory, ethical, and social context of the AI field from which it derives.

This report consists of two parts. The first part describes the historical development of the specific AI techniques used in Coursework 1, including machine learning methods such as regression and ensemble-based approaches, and search-based routing heuristics based in traditional optimization analysis, before examining the conceptual and theoretical foundations that explain how these methods work and situating the project within the larger paradigms of AI. It then connects this lineage to the project's contemporary implementation, assessing which historical accomplishments were most significant in establishing the specific method selected for this specific problem domain that was worked on.

The second part talks about the ethical, legal, and social aspects of the system. This includes a critical evaluation of its potential benefits and drawbacks, the degree of human control it retains or eliminates, and the transparency of its decision-making, especially considering the trade-off seen in Coursework 1 between a more accurate but less interpretable model along with a more transparent but less accurate one. The analysis then considers questions of equity, diversity, and inclusion, examining whether the system's design may disadvantage particular stations or areas, before analyzing its environmental impact and long-term sustainability, and concluding with an analysis of the legal and regulatory aspects relevant to a system based on real-world user trip data.

Throughout this report, the discussion is based on examples from the Coursework 1 implementation, which functions as a case study for exploring these broader theoretical, ethical, and social concerns rather than considering them conceptually.

Part A – Critical and Historical Analysis of the AI Application

Historical Context of the AI Techniques Employed

The machine learning-based forecasting and search-based routing approaches utilized in Module 1 date back to the early days of artificial intelligence, when a study argued that learning and reasoning could be defined exactly as in a human, to allow a machine to simulate them (McCarthy et al., 1955). This idea defined machine learning and automated search as compatible aspects of the relevant field from the start.

The forecasting component has a long history, dating back to a study on machine learning in game of checkers, which invented the phrase and demonstrated that a program could enhance its performance via experience, directly supporting the Linear Regression and Random Forest models utilized in this study (Samuel, 1959). Random Forest was specifically discussed, and decision tree methods and bagging techniques were also investigated, demonstrating that combining multiple models trained on reprocessed data reduces prediction error (Breiman et al., 1984; Breiman, 1996). Then in a 2001 study, it was shown that, these were unified into Random Forests (Breiman, 2001), the method that ultimately performed best in this project. A study, conducted in 2012, documented this progression from early formulation to later refinements, establishing the algorithm's continuing importance within current machine learning practice and demonstrating that this area is still an active area of development instead of a closed topic (Ali et al., 2012).

The next technique, routing, goes back and flows through the Travelling Salesman Problem, one of the most thoroughly studied problems in operations research. The Nearest-Neighbor heuristic utilized in this area was formally published, and another study proved early large-scale computing route optimization (Flood, 1956; Dantzig et al., 1954). Another study later introduced 2-opt local search (Croes, 1958), a refining technique employed in the routing problem, which was explored during this specific project's development.

These two categories, which are, heuristic search over complex problems and statistical learning from data, represent two of AI's most ancient and well-established paradigms. The hybrid design of this project directly inherited both of these lineages, which are still being actively studied today.

Conceptual Foundations and AI Paradigms

AI has traditionally been understood through two broad, but often opposing frameworks. One is, the symbolic paradigm, formally defined in a 1976 study "physical symbol system hypothesis", expresses information clearly as symbols and rules and solves problems by searching the space of possible states (Newell & Simon, 1976). This clearly describes the functioning of this project's routing algorithms, in which each is a search process that uses explicit rules to navigate the space of possible station-visiting orders. The other is the connectionist paradigm, discovered in a 1958 study, which instead learns statistical patterns from data using no explicit rules (Rosenblatt, 1958), similar to how Random Forest and Linear Regression learn demand patterns from historical trips.

This project's hybrid Search and Learning Architecture places it at the intersection of the two paradigms, rather than committing to one exclusively. This is not an unusual combination, and it reflects a developing field of AI study known as neuro-symbolic AI (Colelough & Regli, 2025), which specifically seeks to combine the data-driven pattern identification of statistical learning alongside the structured, explainable reasoning of symbolic search.

A paper published in 2022 refined the topic further, identifying several distinct ways to combine learning and symbolic reasoning, ranging from using one to guide the other to using them as separate but cooperating stages (Kautz, 2022), with the latter method closely matching the architecture of this project, in which a learned forecast feeds into an explicit search process rather than the two being fused into a single model.

To summarize, this study is more than just "using two techniques", but a reflection of a well-known, actively researched hybrid paradigm, with the stated purpose of integrating the flexibility of learning with the transparency of symbolic reasoning. These trade-offs were previously discussed in the project's Critical Discussion part of the technical report.

Contemporary Applications and Impact

The historical developments discussed in the previous chapter did not remain theoretical. They now undergo active, real-world research into exactly the problem this project addresses. Random Forest, from the papers (Breiman et al., 1984; Breiman, 1996), talking on bagging and ensemble decision trees, has become a standard tool in contemporary bike-share demand forecasting.

One study (Fan et al., 2019) combined a random-forest forecasting method with an optimization-based routing method to address unserved demand in bike-share rebalancing, resulting in a nearly identical architecture to the one used in this project. This demonstrates that this particular historical lineage, ensemble learning utilized in relation to demand prediction, has a direct current study, rather than being a random choice.

Similarly, the Nearest-Neighbor heuristic described in the paper (Flood, 1956) remains fundamental to contemporary vehicle routing research. Two other studies (Dell'Amico et al., 2014) and (Schuijbroek et al., 2017) explicitly describe bike-share rebalancing as a vehicle routing problem, building on the same heuristic search tradition, while another study (Liu et al., 2016), applies these concepts directly to New York City's Citi Bike system, which is the same network data used in this project.

In terms of historical breakthroughs, two were particularly noteworthy for this problem domain. In a research (Breiman, 2001), the ensemble technique was important since bike-share demand data is really noisy and station-specific, exactly the type of irregular pattern that a single, simpler model suffers with, and this project's own results verified this. The paper (Flood, 1956), also demonstrates that the Nearest-Neighbor heuristic was important because rebalancing routing needs to be resolved quickly and repeatedly, hour after hour, making a fast, simple heuristic more practical than an exhaustive search, even though this project also confirmed that its solution was, in fact, optimal for the scale involved.

As a result, the transition from fundamental research to present practice is not indirect. This work is part of an ongoing line of research that applies decades-old algorithmic principles to a modern, data-rich operational problem, rather than simply recycling past techniques for an unrelated goal.

Part B – Ethical, Legal, and Social Analysis

Ethical Implications

In terms of possible benefits, identifying station imbalance before it arises and efficiently routing a vehicle to fix it could reduce the number of failed trips for riders while also saving the operator their fuel and staff time. According to a 2019 study (Floridi & Cowls, 2019), based on the paradigm of beneficence and nonmaleficence, the AI system was intended to benefit, but there was also the potential for harm. A faulty forecast could direct a vehicle to the incorrect station, leaving an extremely urgent station unattended while resources are diverted elsewhere. This is a realistic challenge given the per-station error variation reported in Coursework 1's technical evaluation, which revealed that the busiest stations were also the least accurately forecasted.

In terms of human control, this study's approach is intended to support decision-making rather than replace it. It generates a prioritized list and a suggested path, but a human operator is still able to override them. This is essential since full automation of routing decisions would remove an experienced operator's ability to factor in local knowledge that the model does not perceive, such as, for example, a temporary road closure or a scheduled local event.

In terms of transparency, it is where this project's own results become directly relevant. In terms of transparency, this project's own findings are directly relevant. Random Forest, the best-performing forecasting model, is far more difficult to interpret than Linear Regression. For example, a station management may not be able to understand why a given prediction was made, but only to know it was. This is the "black box" scenario, which is identified as a primary ethical concern in deployed algorithmic systems, because a choice that cannot be explained is therefore impossible to justify or alter when incorrect (Mittelstadt et al., 2016). The routing algorithms are more visible in comparison, as each visiting decision can be tracked back to an explicit rule, demonstrating a genuine trade-off between the accuracy and clarity throughout the two sections of this project.

In terms of fairness, the only downside is, the system's primary focus on the twenty busiest stations, as already explained in the technical report, runs the risk of deprioritizing lower-traffic stations, regardless of who uses them.

Equity, Diversity, and Inclusion (EDI)

When considering bias in this project, it is more likely to come from data and design scope than from the algorithms directly. The system was trained and tested exclusively on the twenty busiest stations in Jersey City and Hoboken networks. This is a defensible engineering decision, justified in Coursework 1 by the need to keep the problem manageable, but it has a substantial fairness consequence, which is, lower-traffic stations, which may disproportionately serve residential neighborhoods less connected to major transit hubs, receive no forecasting or rebalancing attention under the current system. A 2016 study described this pattern precisely, in which a system's design choices, without clear aim, have a differential impact on who benefits from them (Barocas & Selbst, 2016).

Discussion about representation is a relevant concern here. The model was trained on historical rider travel patterns, which are influenced by who already has easy access to a nearby station, who can afford a membership, and who works a job with fixed travel times. Riders with irregular schedules, or those already disadvantaged by the station network's location, are not sufficiently represented in the data used to train the model, implying that the system's concept of "normal demand" reflects existing usage patterns instead of a true underlying need. This is the type of representational bias that has been identified as a general concern in machine learning systems trained on non-representative data, including in their own domain (Buolamwini & Gebru, 2018).

This project did not include any explicit design considerations for accessibility. It is important to note that the system optimizes for vehicle routing efficiency during the rebalancing process, not whether the resulting bike availability improves access for riders with mobility issues, who may rely more heavily on a bike becoming available at a specific, nearby station rather than a further one.

To address these gaps, practical strategies need to be included, such as, broadening the scope beyond the twenty busiest stations, weighting the priority score to accommodate stations serving underpopulated areas rather than just raw traffic volume, and auditing predictions on a regular basis to look for systematically underutilized stations.

Sustainability and Environmental Impact

In terms of energy footprint, this project's footprint is minimal by AI standards, but not zero, and it is based on examination rather than assumption. Training Random Forest, the best-performing forecasting model, required fitting 100 decision trees across approximately six months of hourly station data, and it is a computation that can be completed in seconds on a standard laptop machine with no heavy GPU or specialized hardware, which are heavily not required as well for this. This stands out sharply with the huge deep learning models that dominate contemporary discussions about AI's environmental impact.

In a 2020 study (Schwartz et al., 2020), the authors established the phrases "Green AI" and "Red AI", resulting in the "Green AI" specifically defining about reporting computing cost with accuracy, while allowing for lighter-weight, adequately accurate approaches. In short, they aim to achieve reasonable performance while actively reducing energy consumption, emission levels, and hardware costs. As a result, models such as Random Forest are not ignored in favor of overly huge models.

Choosing Random Forest over an arguably more computationally costly approach, such as a deep neural network, when it has been shown to provide no accuracy benefit for this problem, is a Green AI-aligned decision, even if it was initially made for performance rather than environmentally friendly reasons. The routing component is similarly computationally lightweight, which evaluates only a few flagged stations per hour rather than the entire network, and also completes in less than a second.

In terms of long-term sustainability and scalability, this system's structure presents a serious limitation. The brute-force optimality test which was used to validate the Distance-only Nearest-Neighbor routing approach at the end of the implementation, scaled proportionally with the number of stations, which was the top 20 heavily trafficked stations. So, if this project was scaled out to the entire Citi Bike network of 100 plus stations rather than the twenty most heavily trafficked stations used here, it would become computationally infeasible. However, the fundamental forecasting and routing approaches scale significantly more gracefully, as Random Forest and Nearest-Neighbor-based routing grow almost linearly with the number of stations involved.

Therefore, to make this system greener while preserving its functionality and performance, the brute-force verification step could be permanently limited to periodic sampling instead of routine use, as well as Random Forest's tree count could be minimized further if evaluation revealed only a marginal accuracy cost, directly following the study's (Schwartz et al., 2020) recommendation to treat computational efficiency as a genuine design criteria, instead of a secondary concern.

Legal and Regulatory Considerations

This project utilizes the authentic Citi Bike trip data according to the NYCBS Data Use Policy, which allows for academic use. The dataset contains zero personal rider information such as names, residual addresses, or payment methods, and only general trip-level station, time, location, and bike type records, making it less vulnerable to data protection laws. Since the data originates from a US network, the applicable framework is US privacy law, and the US currently lacks a single federal privacy law.

New Jersey explicitly passed the New Jersey Data Privacy Act, which came into effect on January 15, 2025, and applies to organizations that process the data of 100,000 or more New Jersey customers, or at least 25,000 consumers if data sales produce revenue. Although the live, rider-level user data utilized in this study is anonymous, aggregated, and nonpersonal, it may be subject to this law.

When it comes to accountability, it is a real concern for a deployed version of this study. To be more specific, if a system-guided routing vehicle missed a stop and then ran empty, liability would not be clearly assigned to the system developer, routing vehicle operator, or the impacted rider. This void is precisely what the EU AI Act (Regulation (EU) 2024/1689) seeks to fill, and it remains a relevant reference point even outside of EU jurisdiction, thereby supporting this study.

In terms of ownership, this project's code belongs to the author, while the underlying trip data is Citi Bike and Lyft's original property and is utilized in this study strictly under their data license.

Conclusion

This report analyzes the Coursework 1 system in its historical, theoretical, and social context. Part A tracked the forecasting along with routing techniques back to their historical origins in early machine learning and operational analysis, explored their conceptual foundations within AI's symbolic and connection-based paradigms, and linked this lineage to the current bike-share research, indicating that the project's hybrid Search and Learning approach reflects an established, actively studied methodology instead of just a random combination of methods.

Part B analyzed the system critically rather than just technically, emphasizing on ethical, legal, and social aspects. It concluded that while this study is a decision-support tool with real potential benefit, it has a few genuine limitations, like an accuracy-transparency drawback in its forecasting model, a scope that currently excludes lower-traffic stations, a modest but non-zero energy footprint, and open legal questions about accountability that current regulation in both the US and EU is still actively working to resolve.

In conclusion, both the coursework demonstrate that a technically proficient AI system is neither a complete nor an inherently useful one. Responsible development demands the type of critical evaluation demonstrated here, with ethical, legal, and social considerations treated as fundamental to the work instead of as an additional consideration.

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